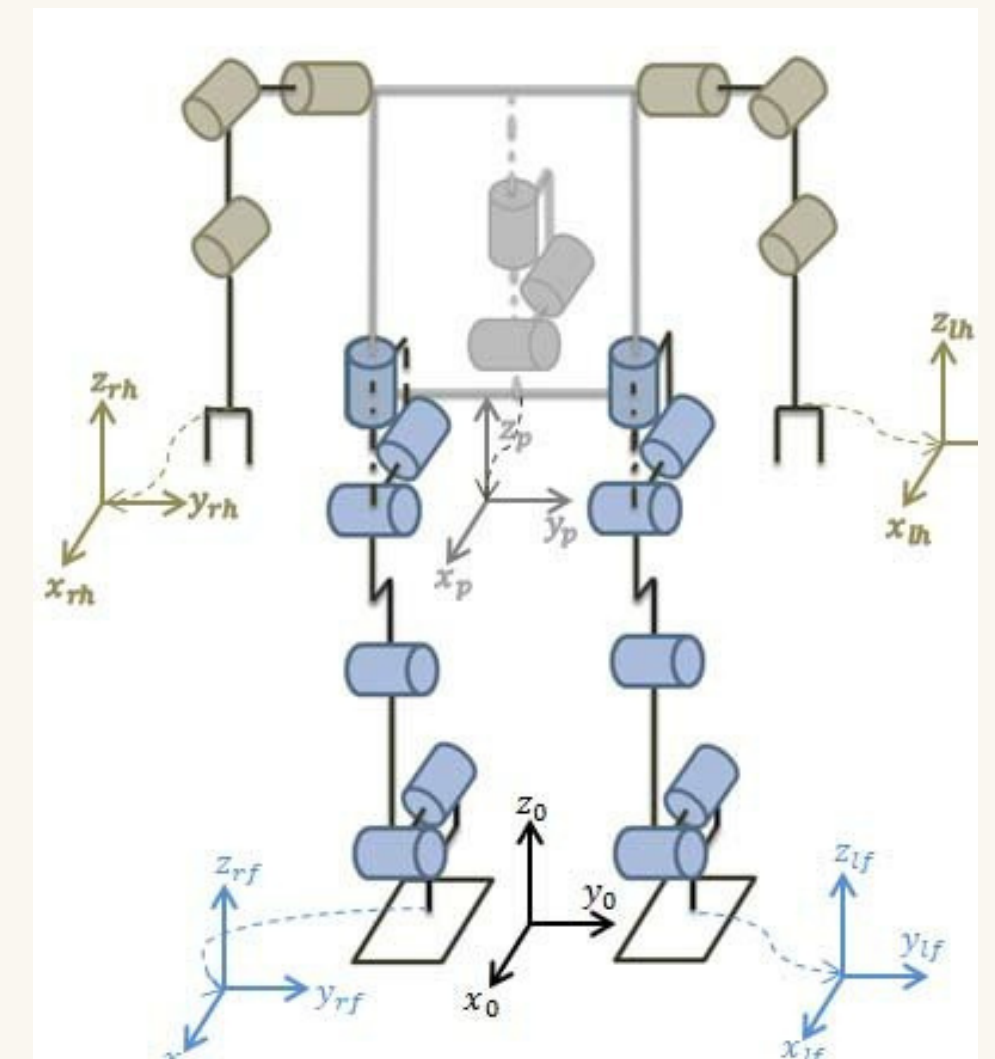
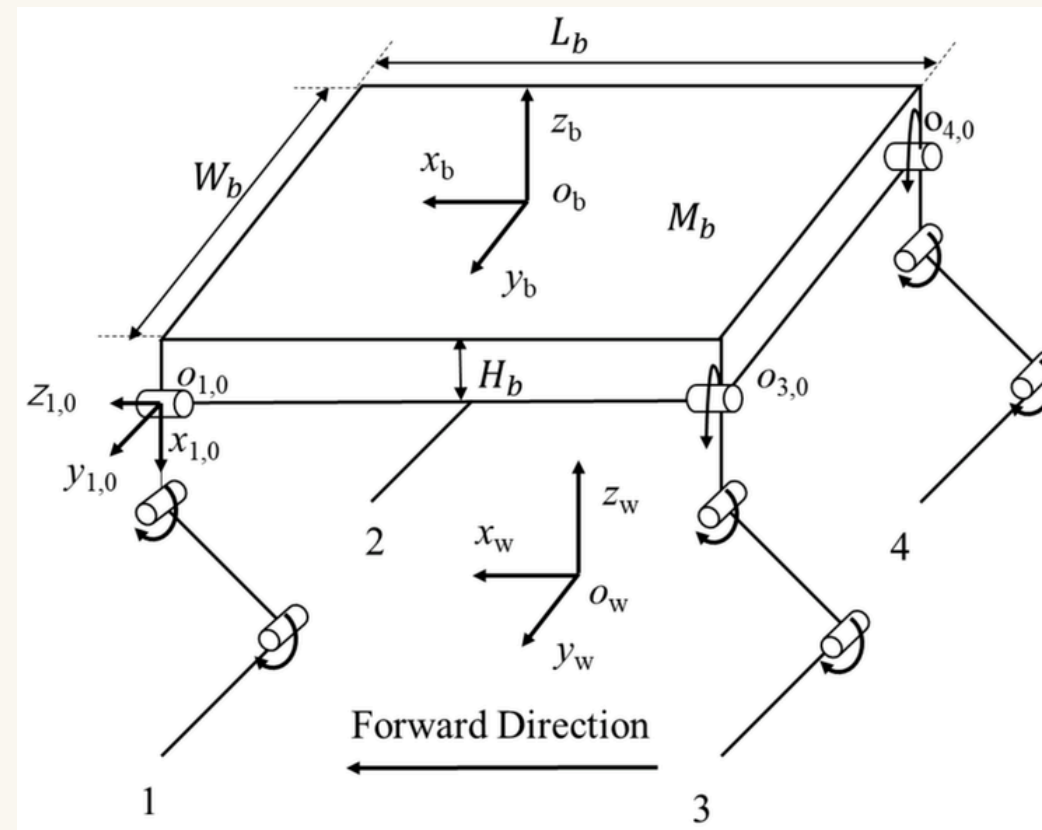
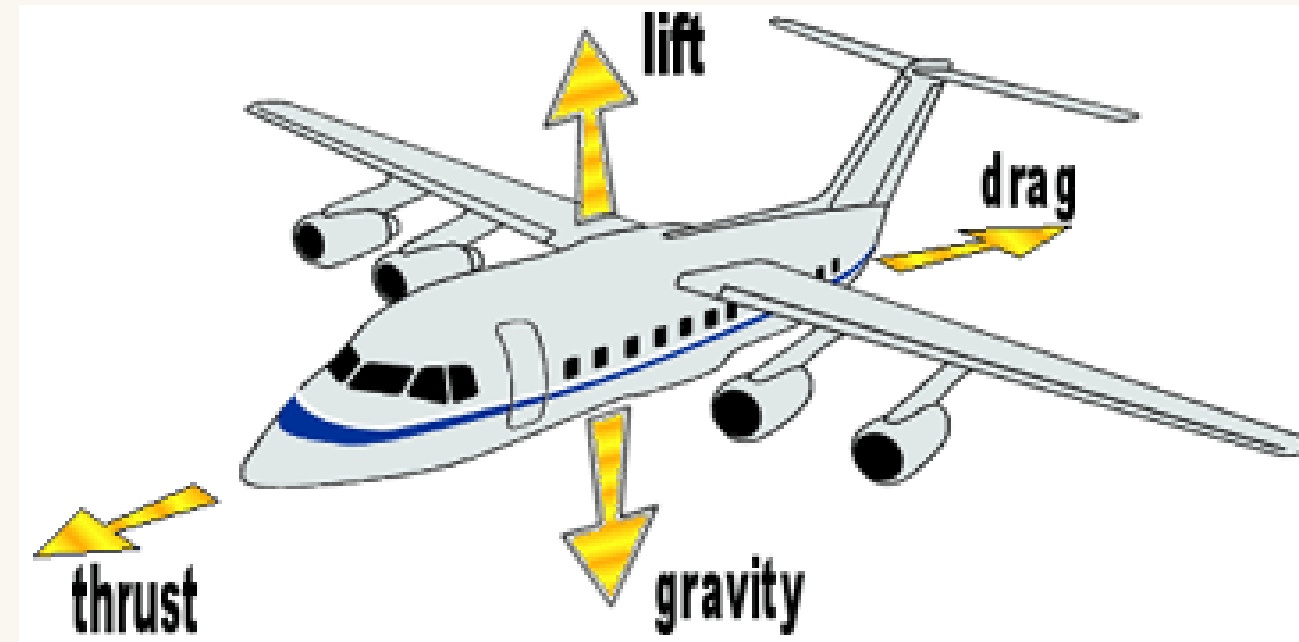
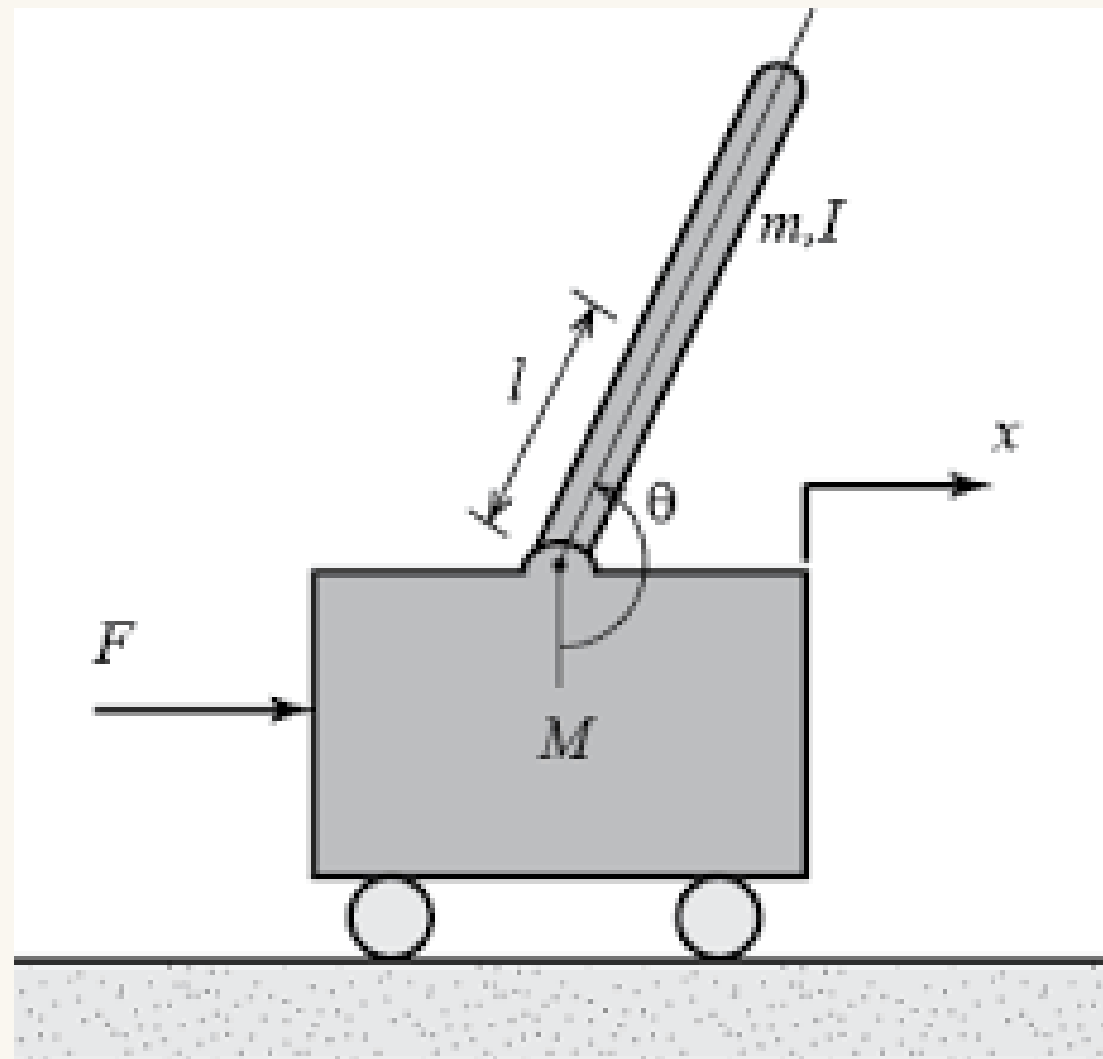


Dynamics and Controls - I

Mathematical Modelling of dynamical systems

Astik Srivastava
Viswas Dinesh

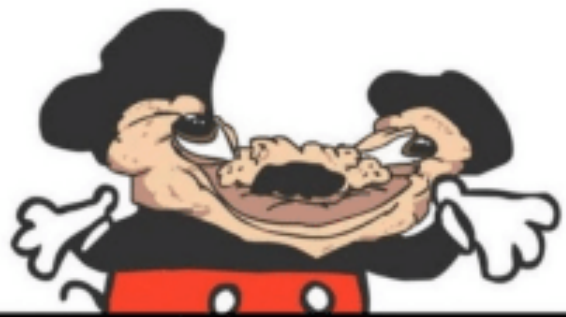
What is a dynamical system?



Linear vs Non Linear analysis

Linear

Accuracy



Usability

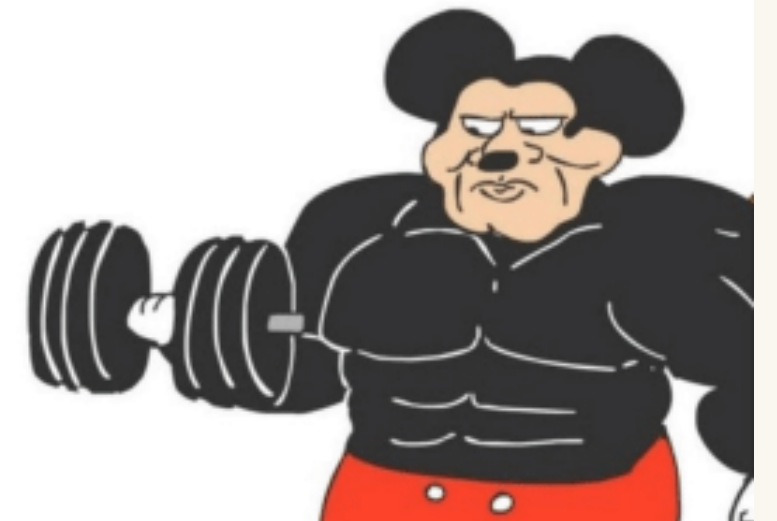


Non-Linear

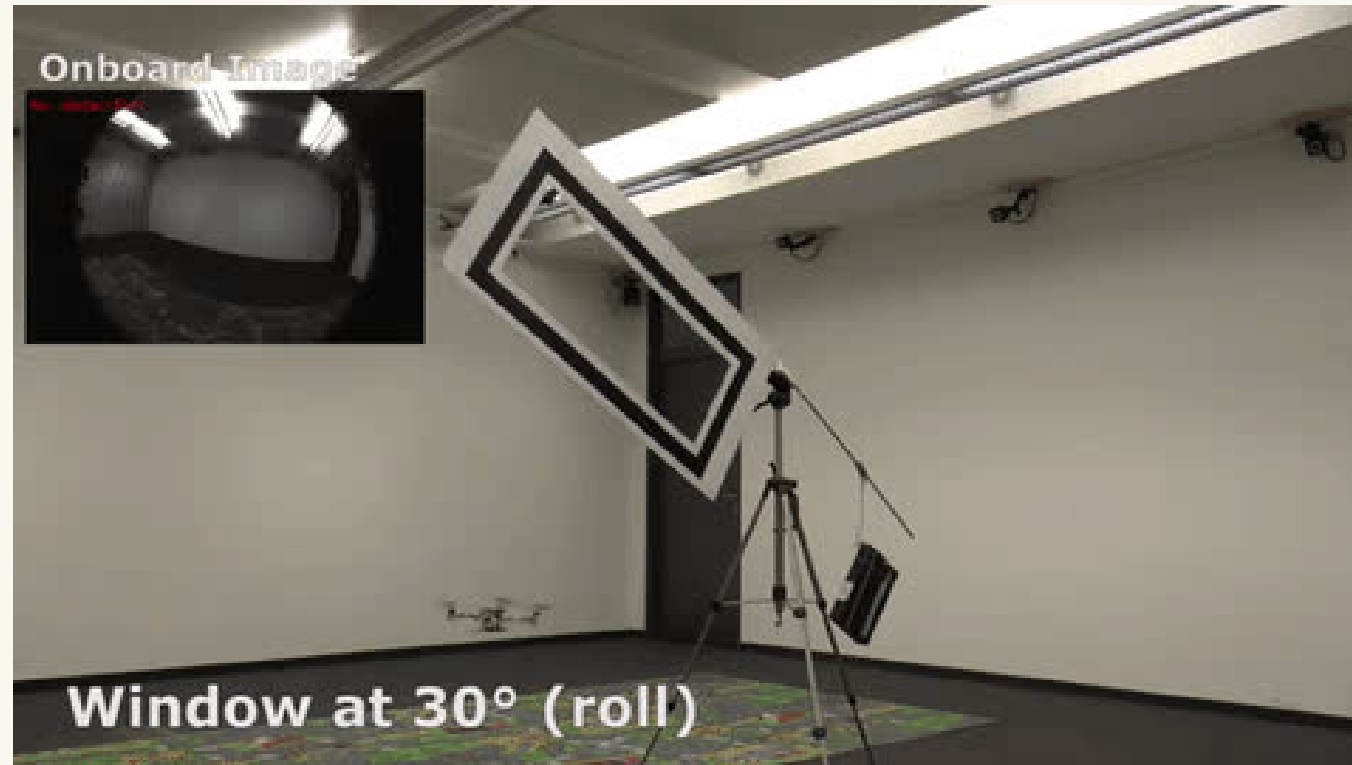
Usability



Accuracy



Why Non Linear analysis ?

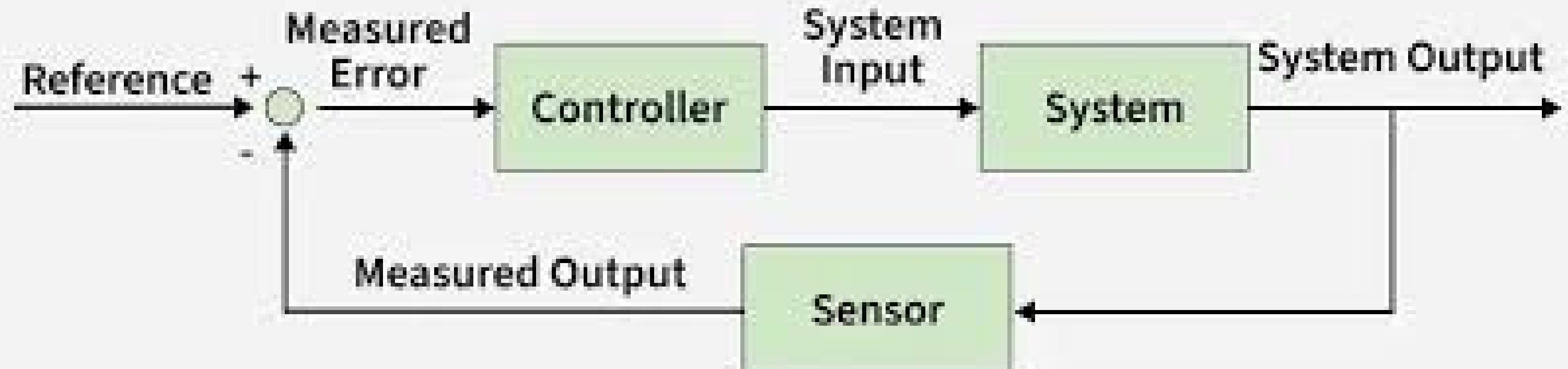


Chad Nonlinear

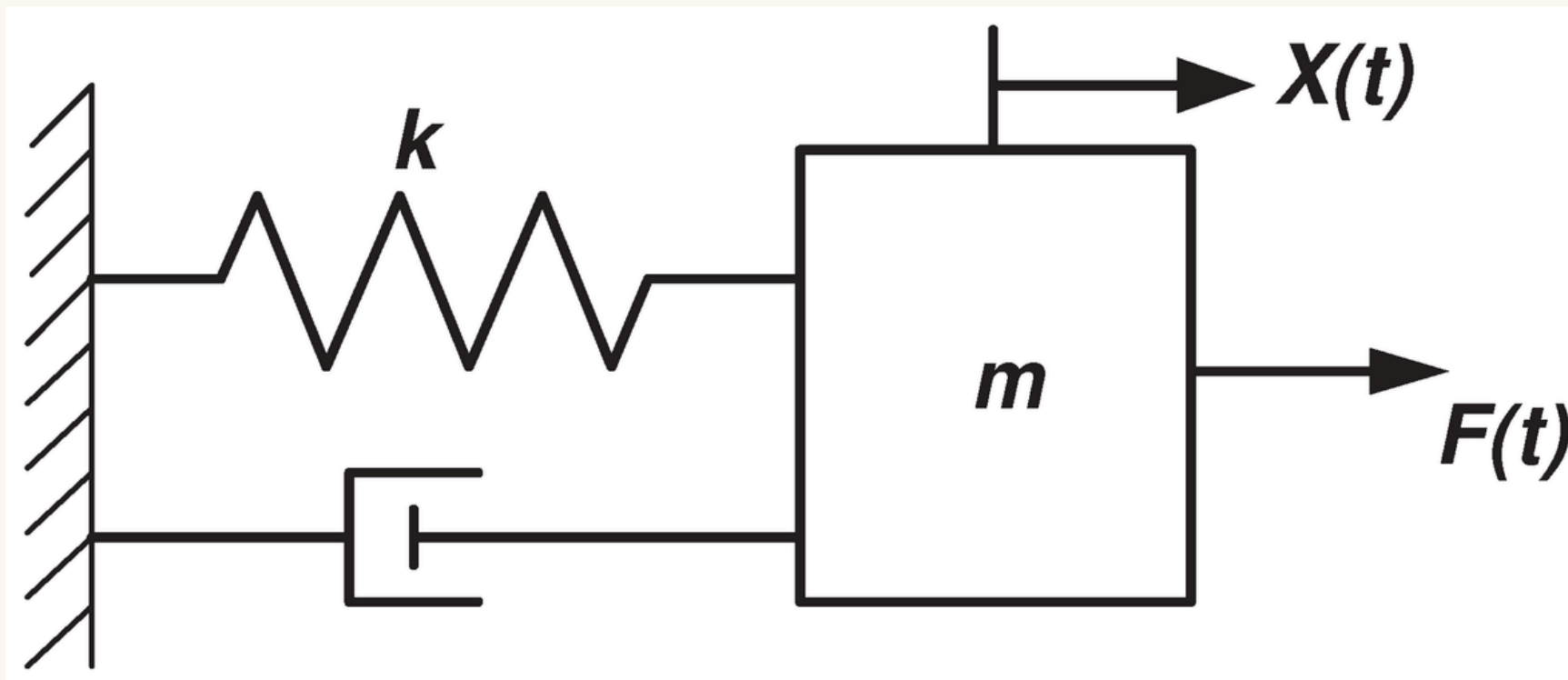
Normie Linear



Linear Dynamical Systems



Linear Systems Modelling

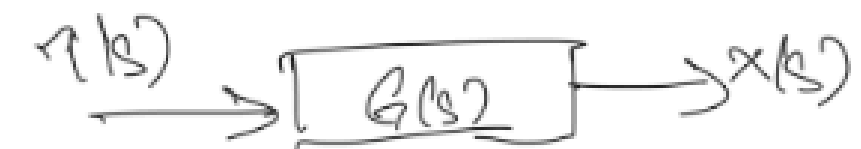


SYSTEM MODELLING

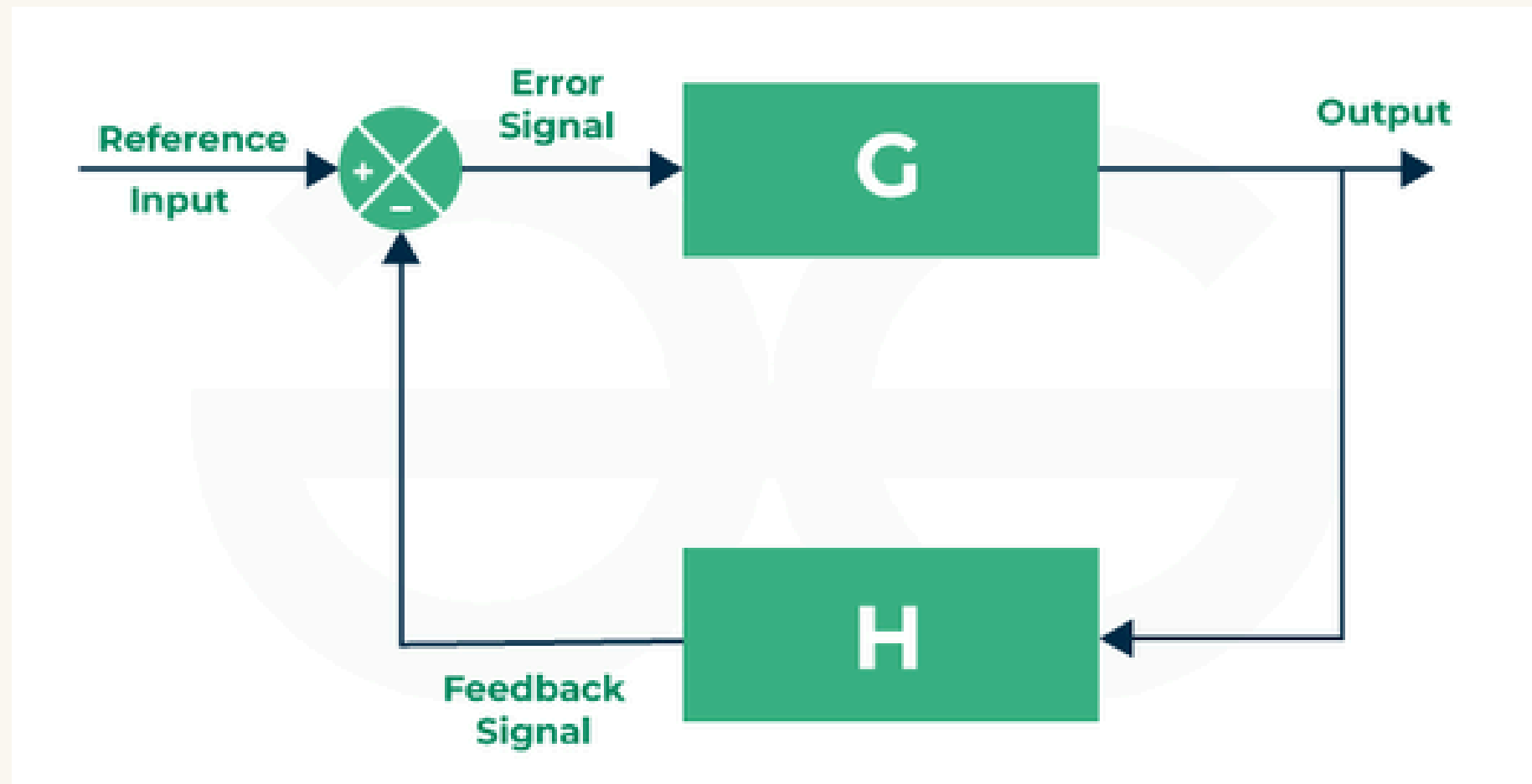
$$m\ddot{x} + b\dot{x} + kx = \tau$$

$$(ms^2 + bs + k)x(s) = \tau(s)$$

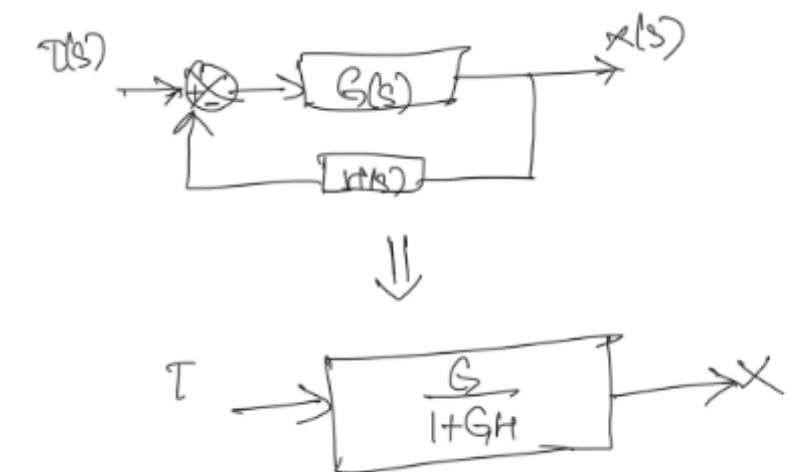
$$G = \frac{x(s)}{\tau(s)} = \frac{1}{ms^2 + bs + k}$$



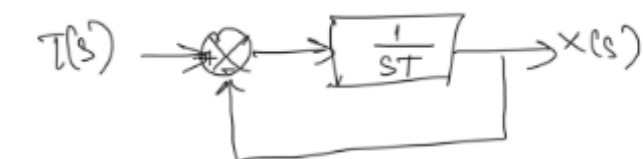
Linear Systems: Feedback control



feedback:



Gets mathematical now



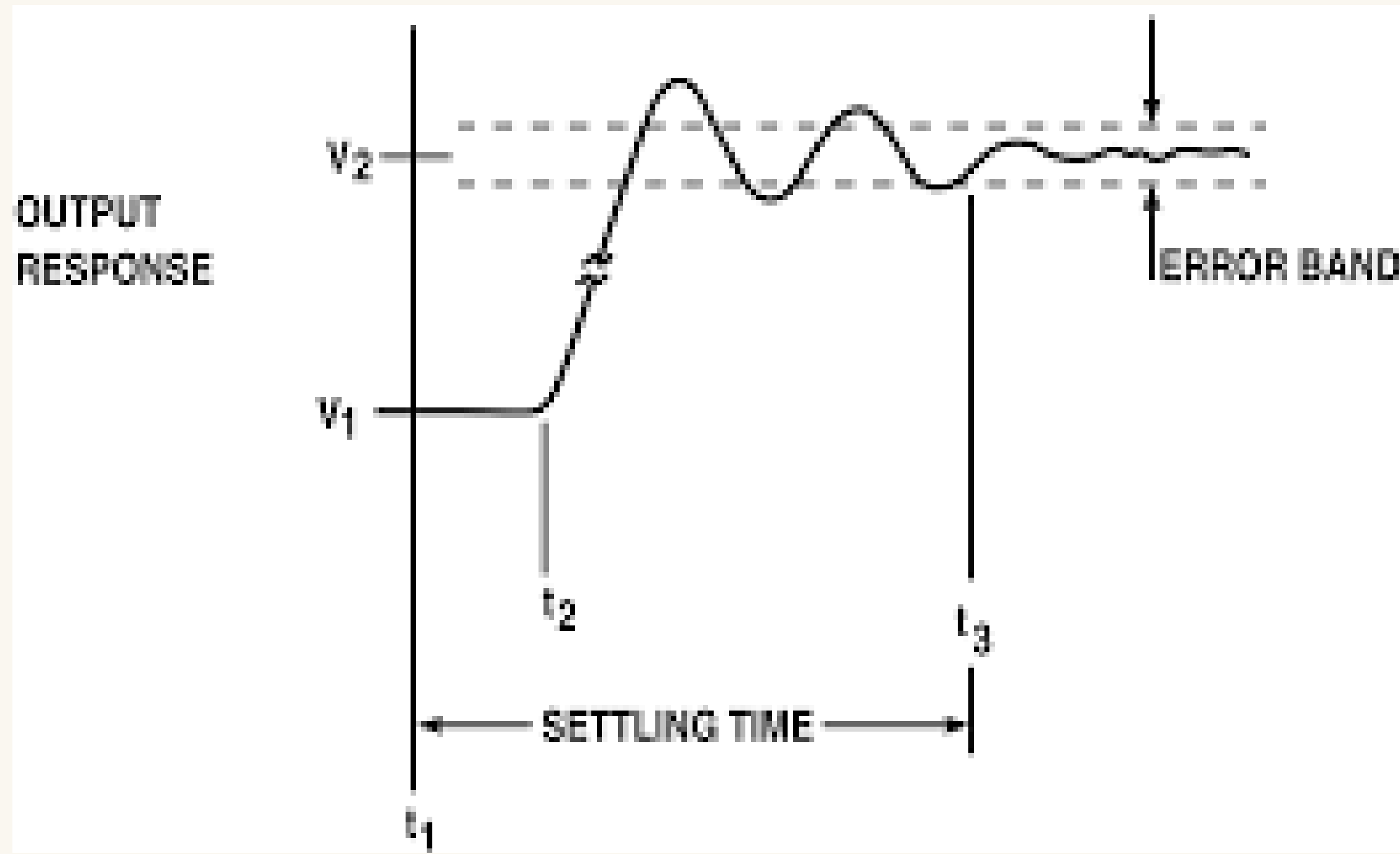
$$TF = \frac{G}{1+G} = \frac{\frac{1}{sT}}{1+\frac{1}{sT}} = \frac{1}{1+sT}$$

$$\Rightarrow X(s) = \left(\frac{1}{1+sT} \right) T(s)$$

Step - response:

$$T(t) = u(t) \xrightarrow{\mathcal{L}} T(s) = \frac{1}{s}$$

Linear Systems: Step Response



Linear Systems: Stability

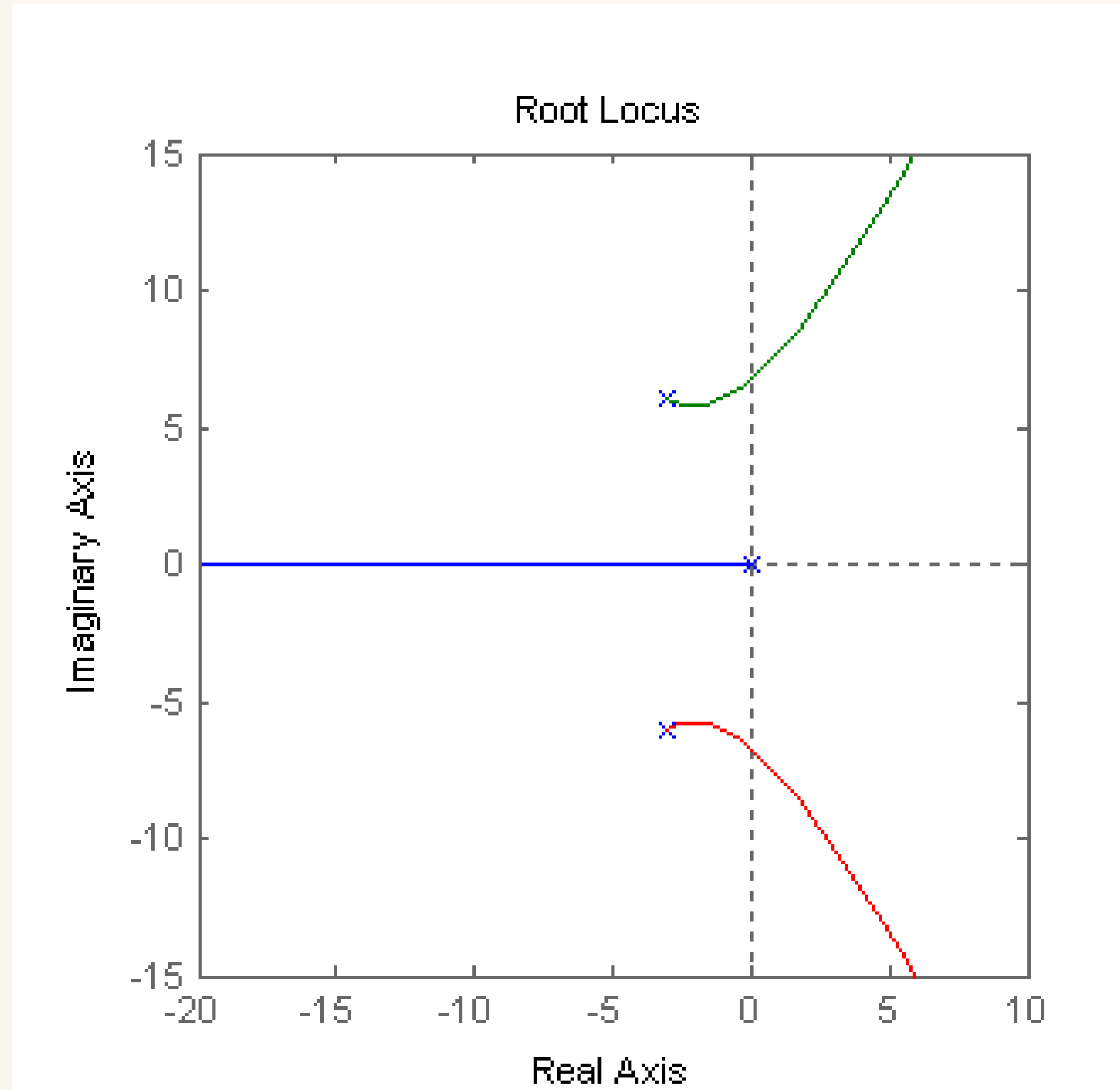


$$T(s) = \frac{N(s)}{D(s)}$$

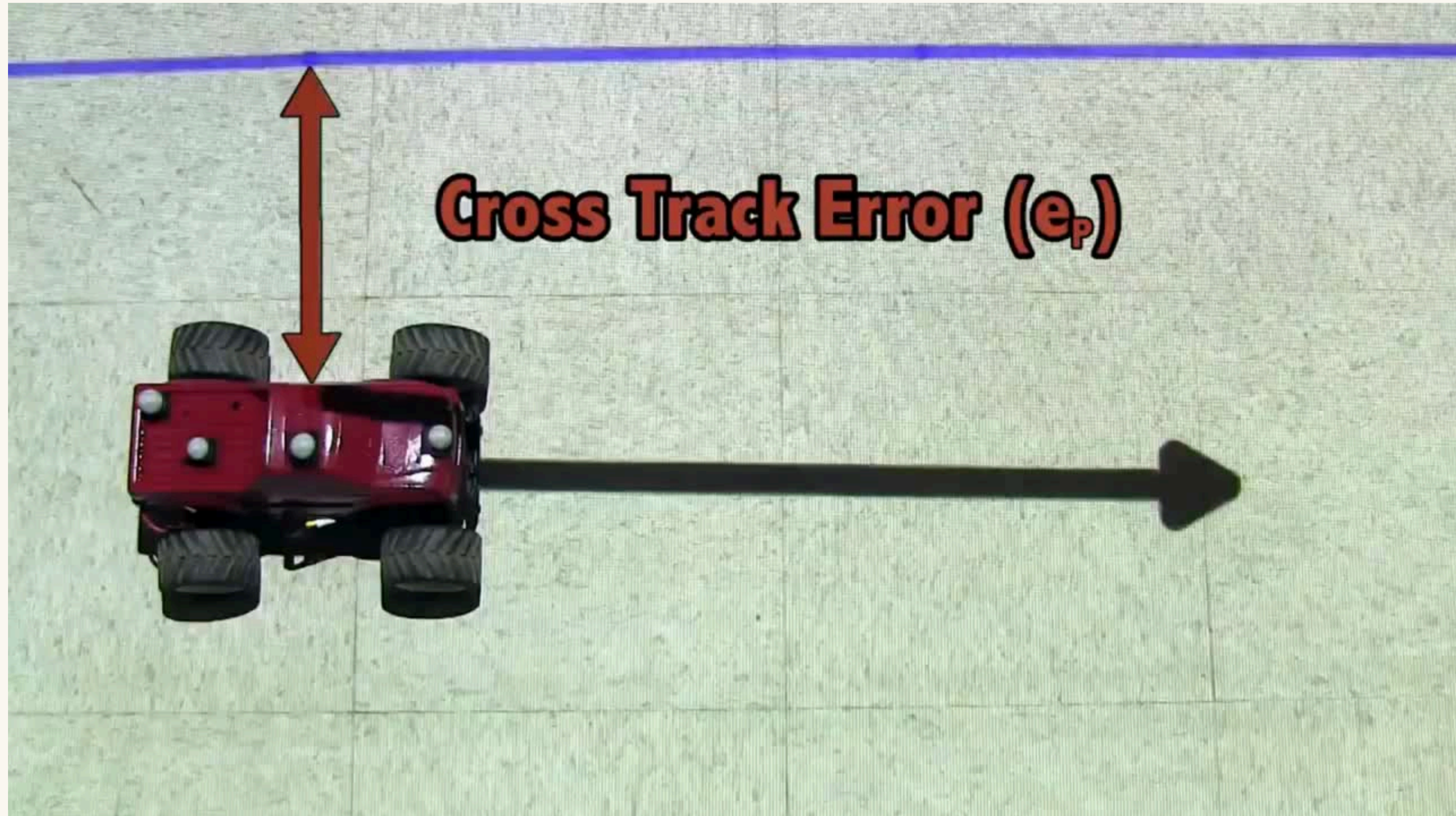
$$N(s_{\text{zeros}}) = 0$$

$$D(s_{\text{poles}}) = 0$$

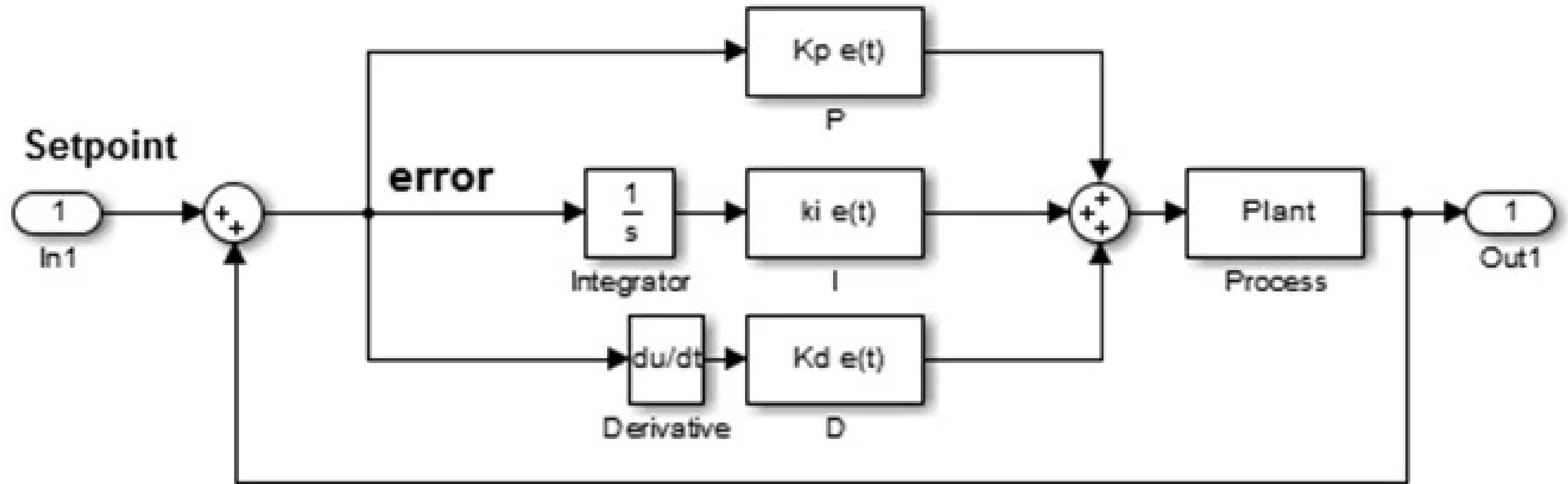
Linear Systems: Root Locus



Linear Systems: Why Root Locus?



Linear Systems: PID controller design



PID Controller